

Multimedia Streaming Architecture

Noviembre, 2016

Audio & Video Delivery

- **Time:**
 - The question is: *¿Live or not live?* «Vivo»
- **Media:**
 - **From:** Analog (e.g. Air)
 - **To:** Digital (e.g. Internet)
- **Authoring:** Copyleft/Copyright
- **Protecting:** Encrypted or at least secured

A&V Digital Processing

- **Hardware & Software:**
 - Open vs. Proprietary
- **Encoding:**
 - Audio / Video (Capture/Record)
 - Analog/Digital (Compression / Lossless / Lossy)
- **Transcode:**
 - Analog <-> Digital (Format)
 - Hardware / CPU / GCPU
- **Transmux:**
 - Digital A&V Format Conversion (a.k.a Transcode)
- **Stream (Digital *Cast):**
 - Server / Client (Playback).
 - Live Stream => Live (Re)Streaming (LS)
 - Recorded File => Video On Demand (VOD)

Digital Streaming Technologies

Streaming Technologies

BENEFIT	LIVE STREAMING	PSEUDO STREAMING	PROGRESSIVE DOWNLOAD
Real-time broadcasts	X		
Long clips	X	X	
Immediate random access to different parts of a movie	X	X	
Downloads entire movie			X
Downloads required part of the movie		X	
Flv is cached on the local system			X
Requires a specialized streaming server	X		
Requires a web server with php		X	
Can be stopped by firewalls	X		
Consistent high quality playback at any connection speed		X	X
Retransmits lost packets		X	X

Streaming Technologies

SCENARIO	LIVE STREAMING	PSEUDO STREAMING	PROGRESSIVE DOWNLOAD
Live broadcasts	X		
Webtv	X	X	
E learning	X	X	
Online rich Media presentation	X	X	
Movie trailers		X	X
Short product training clip		X	X
Offline product (cd, dvd)			X

Streaming Protocols

– Internet Based (IP):

- **UDP (Not Connection Oriented / Datagram)**
 - **Real Time Streaming Protocol (RTSP)** is a network control protocol designed by RealNetworks (1996) for use in entertainment and communications systems to establishing and controlling media sessions between end points
- **TCP (Connection Oriented / Packet)**
 - **RTMP (Real Time Messaging Protocol)** created by Macromedia (Adobe) for Internet stream of Flash Video (FLV ~ H.263 + MP3). Released for public use in 2005
 - **Web Real Time Communication (WebRTC)** new project for native zero install any browser open and free video streaming standart for Internet, started in 2011. Supported by all major browsers Chrome, Firefox, Opera, Android & iOS
- **Others:**
 - **MPEG-2 Transport Stream (MPEG-TS)** is a standard digital container format for transmission and storage of audio, video, and Program and System Information Protocol (PSIP) data.[3] It is used in broadcast systems such as Digital Video Broadcasting (DVB), ATSC and IPTV

Streaming Protocols

— Web Based (HTTP):

- **Progressive Download (Pseudostreaming):** web server based method for seek and buffering, Apache/Lighthttpd (mod_[h264|flvx]), Nginx (http_[mp4|flvx]_module, Nimble Streamer and many others).
 - **HTML 5:** Most of browsers are comming to support open & standart «video audio» combos: WAVE PCM, MP4 H.264 (AAC o MP3), WebM, Ogg Theora Vorbis, Ogg Opus.
- **HTTP Live Streaming (HLS)** created by Apple for streaming audio to its iPhone, iPad, Mac, Apple TV devices (also PC) since 2009.
 - Apple VOD formats takes video file from MPEG-4 / QuickTime MOV to H.264 Compression and audio files to AAC or MP3 compression.
 - Apple LS formats uses MPEG-2 transport streams carrying H.264 video, AAC or MP3 audio.
- **Dynamic Adaptive Streaming over HTTP (mpeg-DASH)** developed by MPEG (Moving Picture Expert Group) is the ISO/IEC 23009-1:2012[14[since April 2012.

Streaming Protocols

— Web Based (HTTP):

- **HTTP Dynamic Streaming (HDS):** created by Adobe as part of Flash (≥ 10.1) using MP4 Fragment Format (ISO/IEC 14496-12:2008) that allows fetch smaller portions of media than HLS. Allows DRM on live or VOD. Improve RTMP Dynamic Streaming (Flash ≥ 6) and HTTP Progressive Download (Flash ≥ 7) from Adobe.
- **Microsoft Smooth Streaming (MSS):** is a pseudo-streaming implementation introduced in 2008 by Microsoft as part of their Silverlight architecture. Microsoft and Akamai works on «www.smoothHD.com».

Streaming Protocols

— Web Based (HTTP):

- **SHOUTcast** cross-platform proprietary freeware for streaming media (audio/MP3) over the Internet developed by Nullsoft but owned by AOL.
- **IceCast:** cross-platform free software (GPLv2) maintained by the Xiph.Org (www.icecast.org) for streaming the following streams: Ogg (Vorbis, Speex, Flac, Theora), Opus, WebM and MP3.

Audio & Video Formats

- They can't be written neither this slide nor this document...

Formats versus Protocols

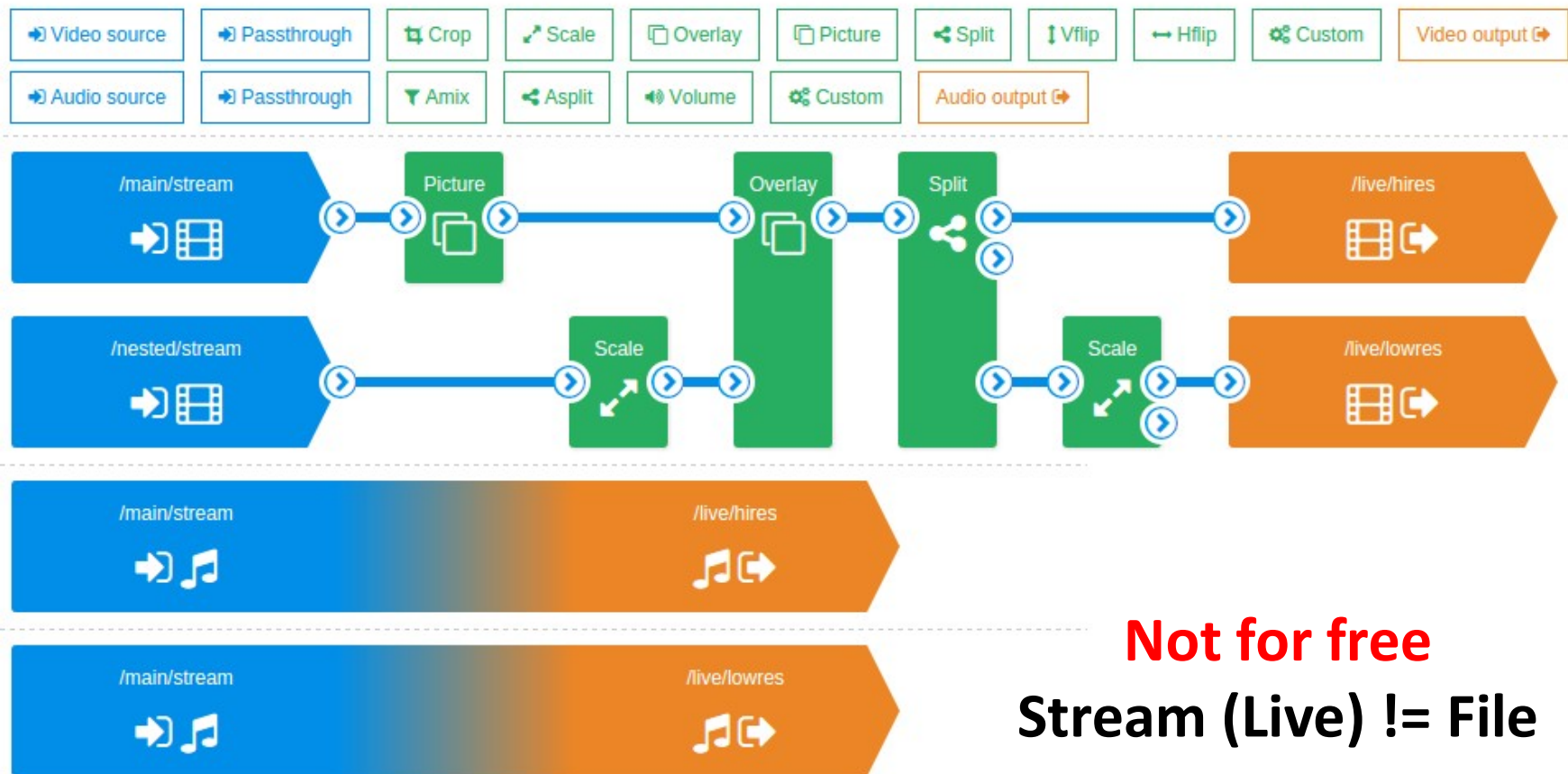
	RTMP input and output	RTSP input and output	MPEG-TS input and output	HLS output only	MPEG-DASH output only	Icecast input and output	SHOUTcast input only
Video							
H.264/AVC	☒	☒	☒	☒	☒		
H.265/HEVC		☒	☒	☒	☒		
MPEG1			☒				
MPEG2			☒				
MPEG4			☒				
VP6	☒						
VP8		☒					
VP9		☒					
Audio							
AAC	☒	☒	☒	☒	☒	☒	☒
MP3	☒	☒	☒	☒	☒	☒	☒
AC3, E-AC3		☒	☒	☒			
Speex	☒						
PCM G.711 (a-law, μ -law)		☒					

H.264 + AAC

Live Transcoder

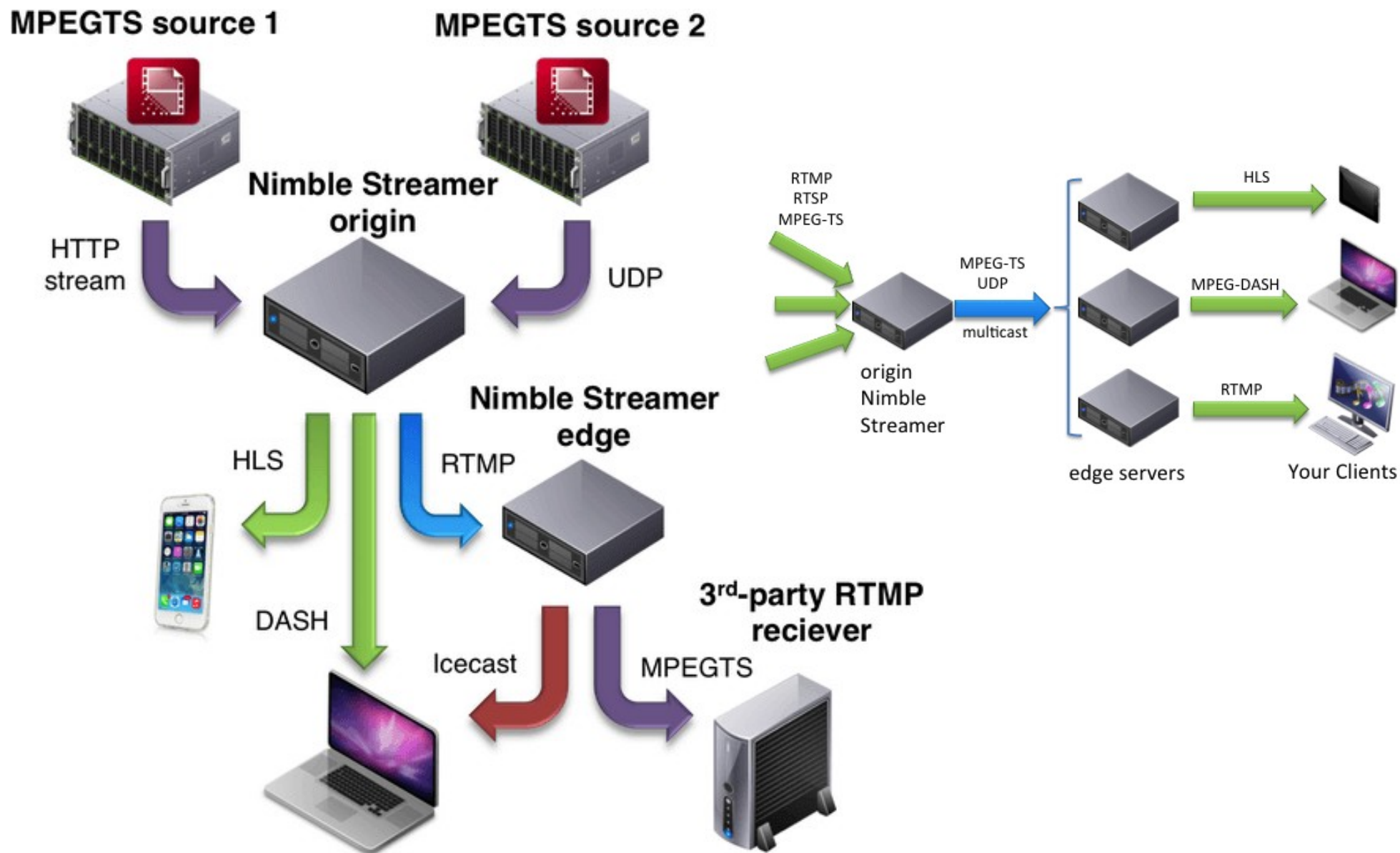
Overlays example

This example puts picture in picture, puts logo on top and creates 2 output resolutions.

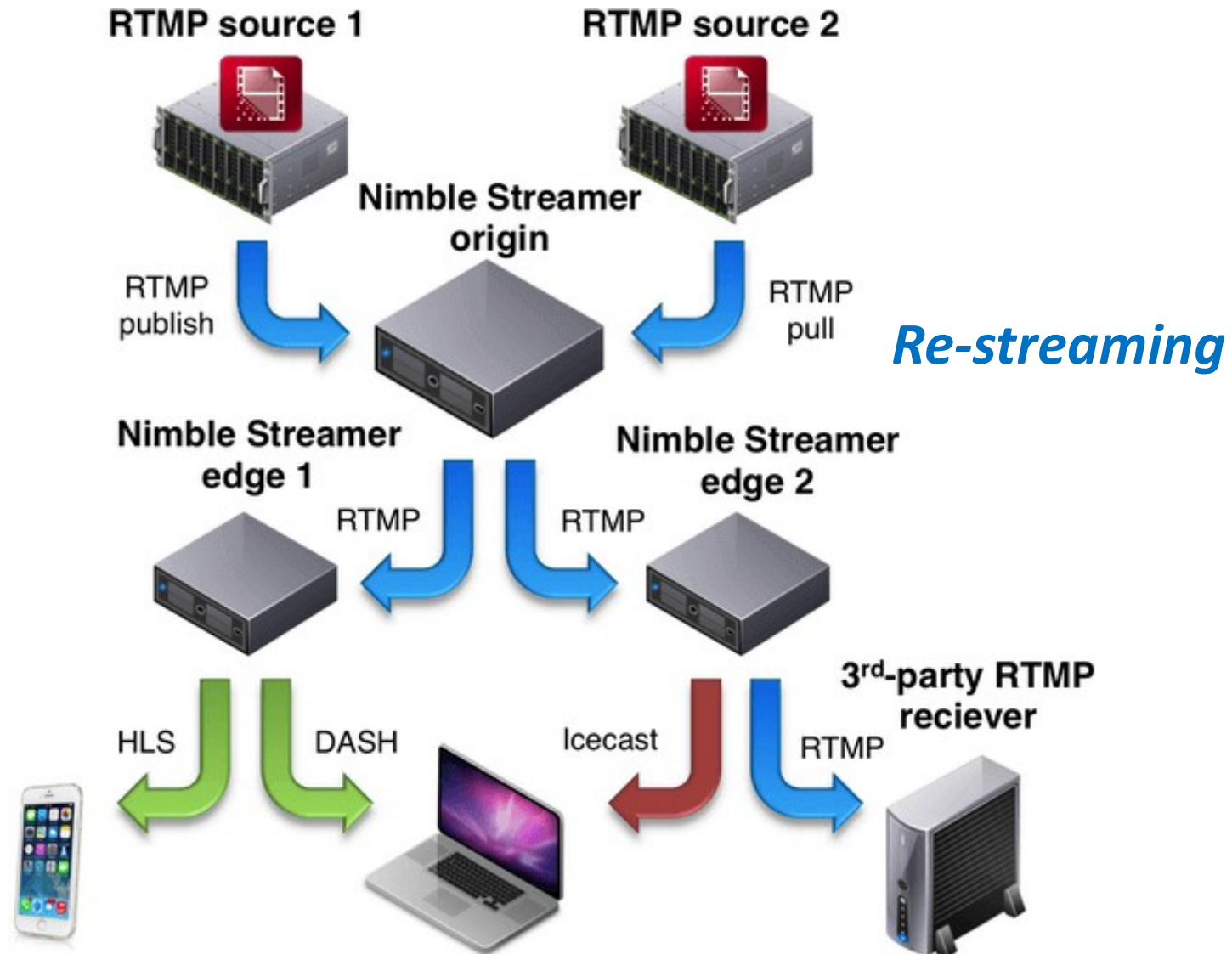


Digital Streaming Architectures

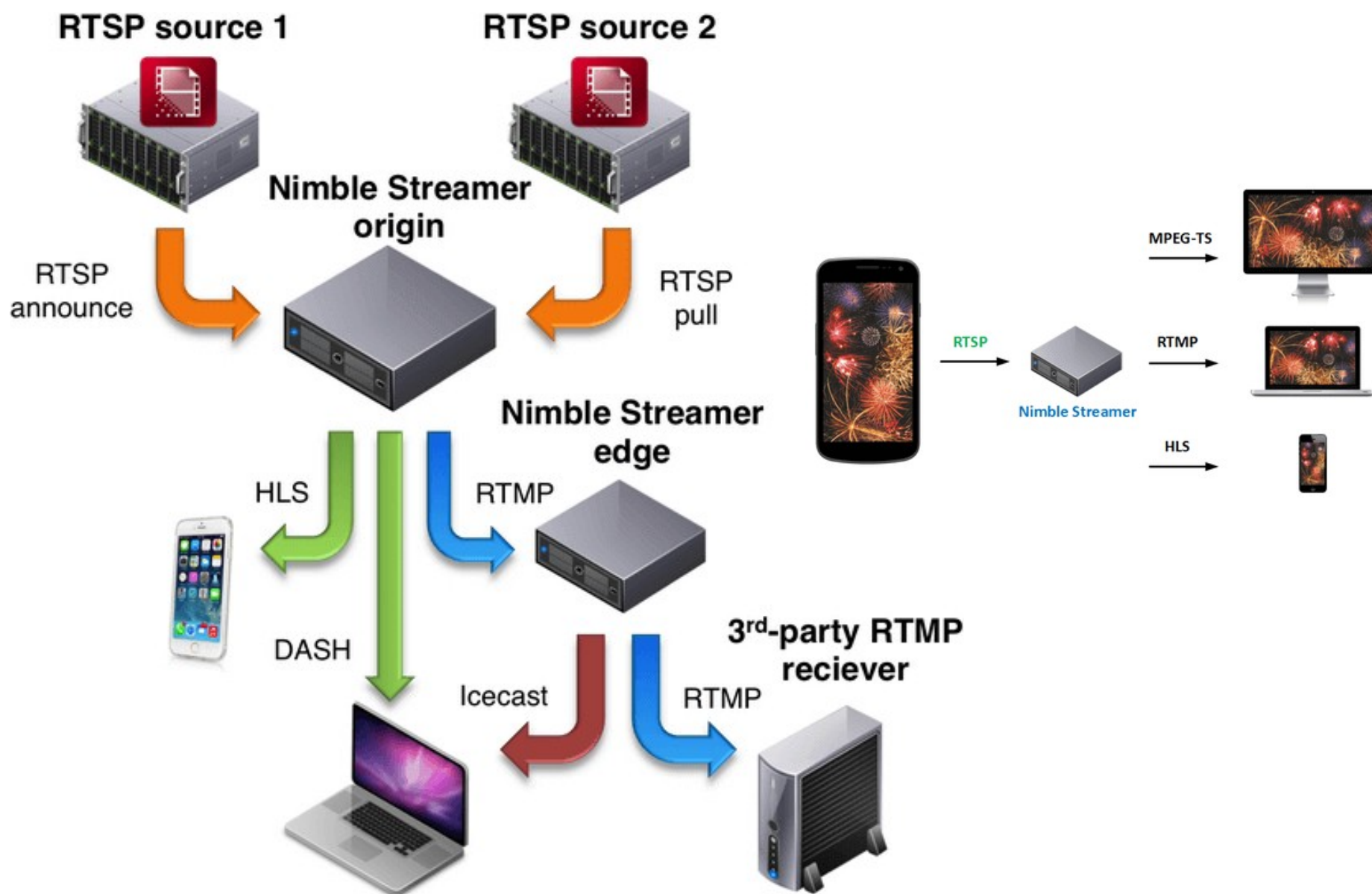
Live Streaming (Transmuxing)



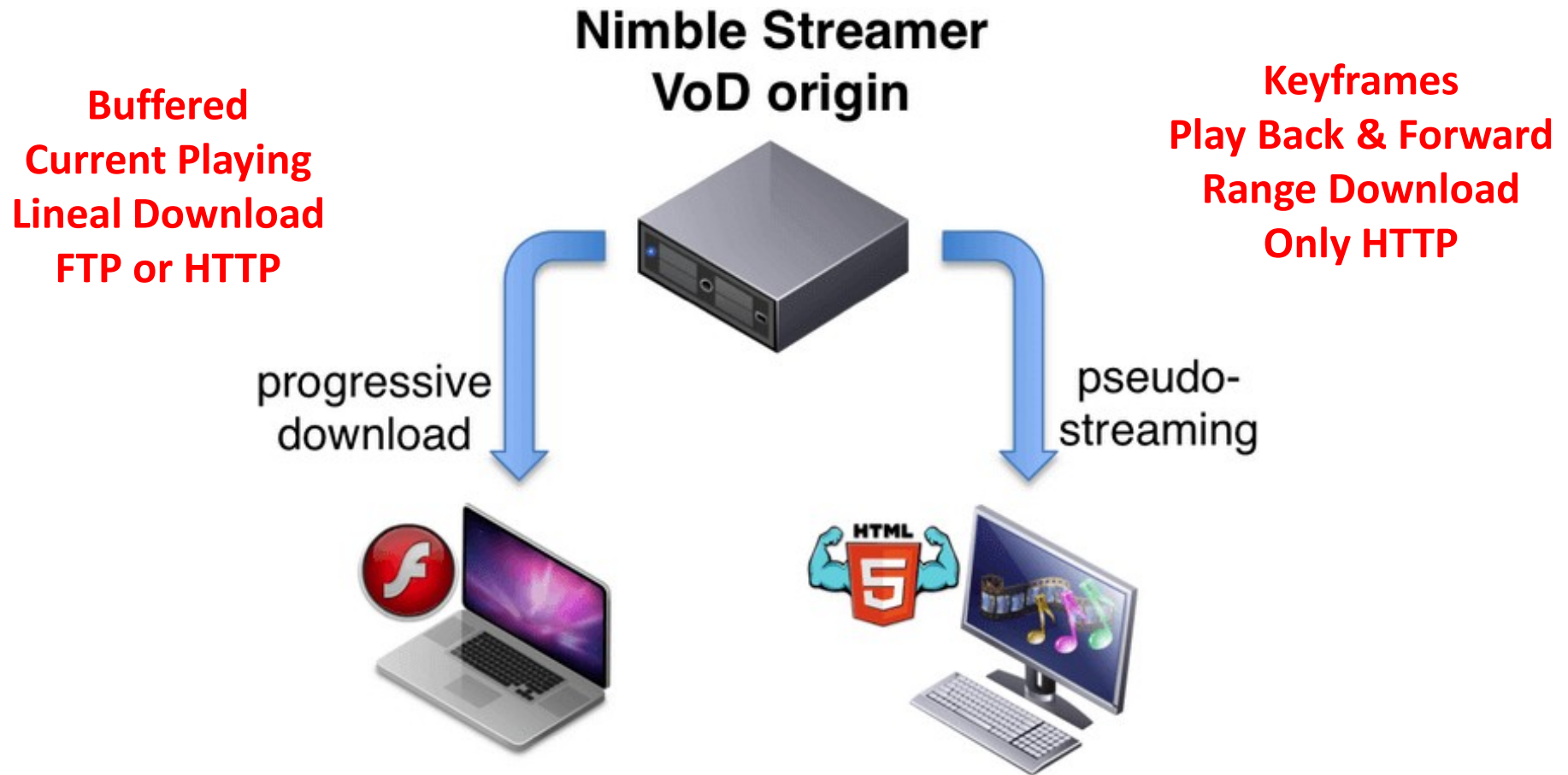
Live Streaming (RTMP)



Live Streaming (RTSP)

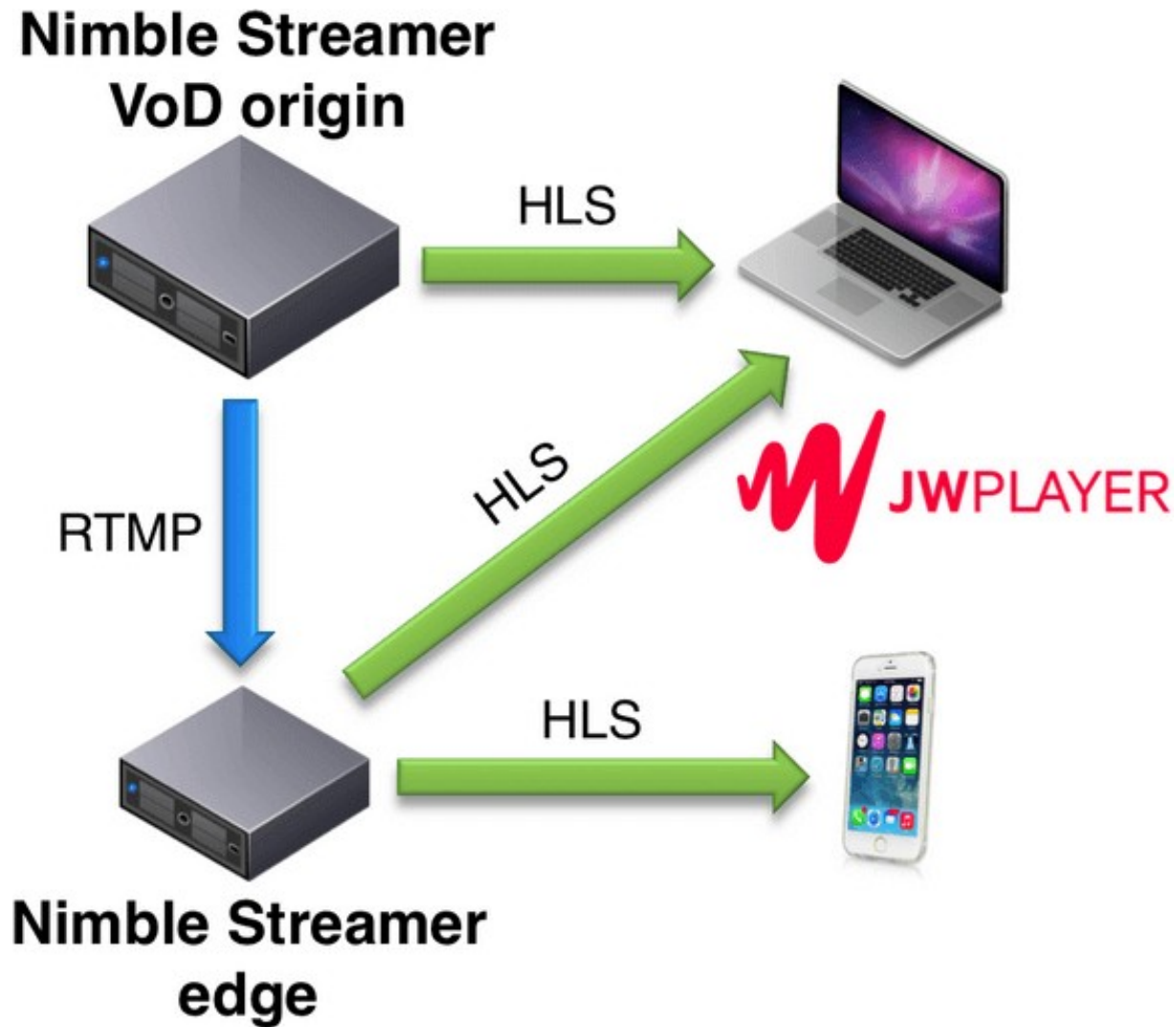


Pseudo Streaming



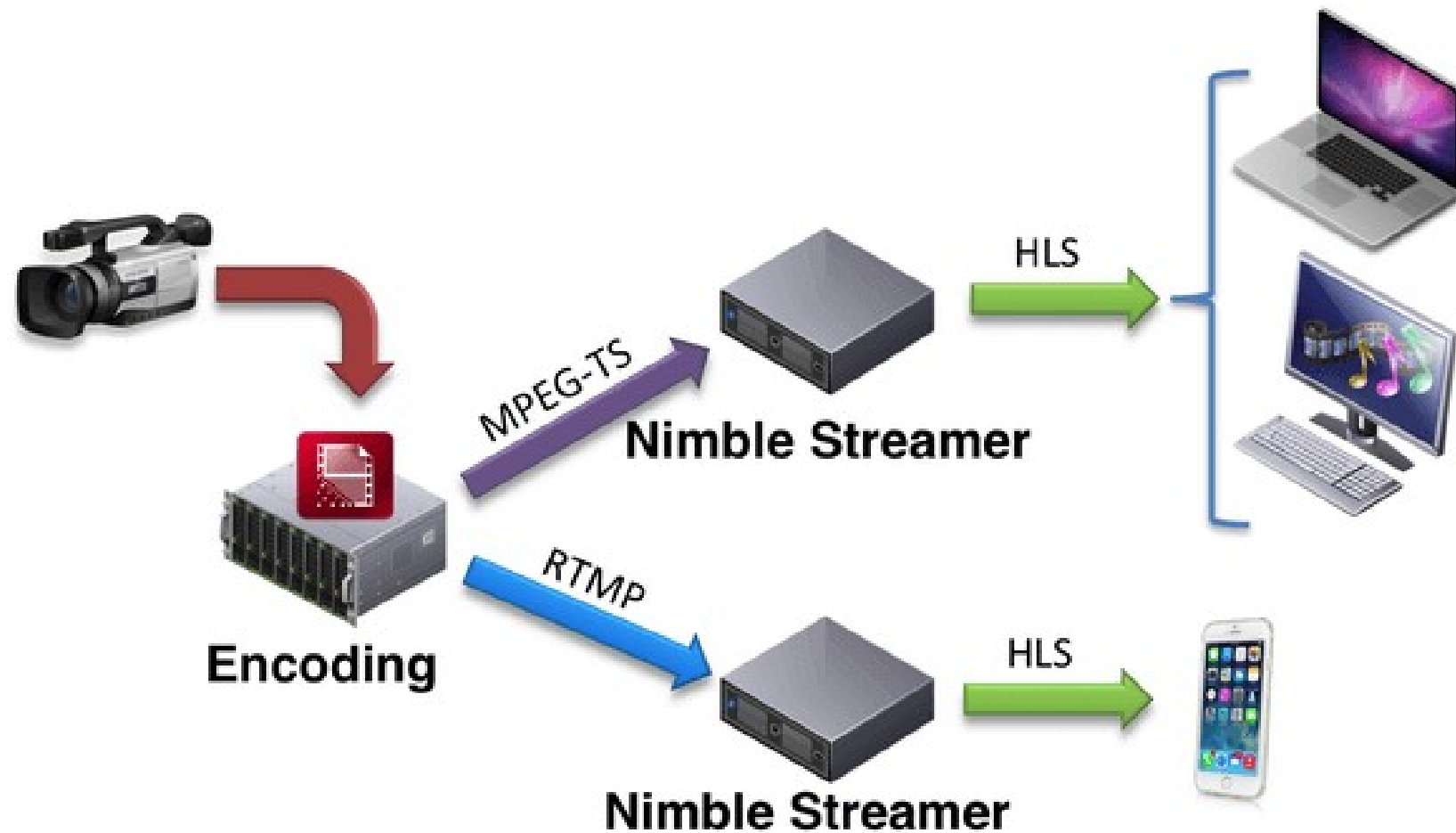
- **File formats:** mp3, mp4, flv, oga, ogv, wav, webm, mkv
- **Optimization:** flash time marks seek, HTTP range header
- **Packet loss retransmission:** which is not available on live streaming

Video On Demand (VOD)



HTTP Live Streaming (HLS)

Live Streaming (HLS)



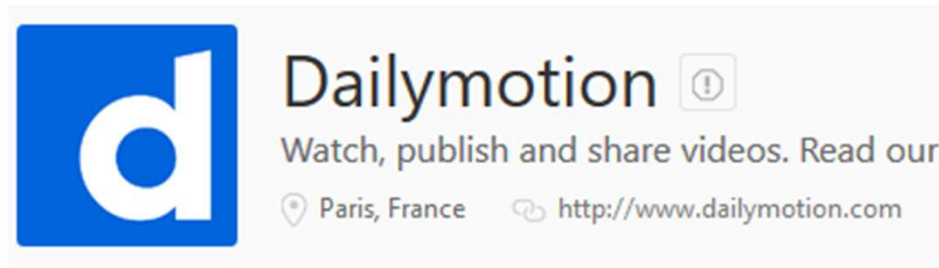
HLS Benefits & Tips

- HLS is not A&V is only the **delivery way**
- Supports (client-driven) **adaptive bitrate**
- **Better experience:**
 - No “partial play”
 - No “Complete download”
- **Delivered over standard HTTP**, so zero firewall blocking policies.
- **Developer and devices easy integration** with simple, text-based manifest format.
- **No proprietary streaming servers** required, so more people streaming & playing back.

HLS Benefits & Tips

- **User are dropping major vendors with proprietary standards** (i.e. Flash).
- **Major browsers no longer support plugins** (zero install, plug & play, security, licensing).
- **HLS video is now supported on multicast** (RAMP Multicast Engine Startup).

HLS Playback Device Support



☒ Pay yearly **20% off** ☐ Pay monthly

PAY AS YOU GO
Best to start with

€0 /mo
Streaming costs
€0.35 /GB
Encoding costs *
€0.05 /min.

Subscribe
You can upgrade later

ECONOMY	PRO	ENTERPRISE
€20 /mo	€119 /mo	€350 /mo
Streaming included 0.3TB	Streaming included 1.5TB	Streaming included 4.5TB
Additional streaming €0.13 /GB	Additional streaming €0.11 /GB	Additional streaming €0.09 /GB
Encoding Included 250 min.	Encoding Included 1000 min.	Encoding Included 2000 min.
Additional encoding * €0.03 /min.	Additional encoding * €0.02 /min.	Additional encoding * €0.02 /min.
Subscribe	Subscribe	Subscribe

FREE	PREMIUM	PLATINUM WITH ADS
\$0	\$299/yr	\$999/yr
START FOR FREE	GET IT NOW	GET IT NOW
25GB Streaming/month 10k video plays/month	250GB Streaming/month 100k video plays/month	500GB Streaming/month 200k video plays/month
Included Features: HTML5 Video Player Android & iOS Mobile SDKs Apple TV Showcase Apps	Free Features Plus: HLS Adaptive Streaming Multi-bitrate HD Encoding AirPlay & Chromecast	Premium Features Plus: HTML5 & Mobile Ad Insertion VAST/VPAID Advertising Certified Google IMA Integration Advanced Ad Scheduling Ad Waterfalling Ads-Enabled Mobile SDKs

- Free
- Plays (Clicks)
- Streams (Data)

HLS Playback Device Support

5 Minutes App Store Limit

Apple Inc. (US) | https://developer.apple.com/library/content/qa/qa1767/_index.html

Buscar

Guides and Sample Code

Developer

Resolving App Store Approval Issues for HTTP Live Streaming

Resolving App Store Approval Issues for HTTP Live Streaming

Q: My Application was rejected with the "HTTP Live Streaming rejection 9.4". What does this mean and where can I find information about correctly authoring my live streams for approval?

A: The Rejection 9.4 – HTTP Live Streaming rejection notice specifically states:

"Video streaming content over a cellular network longer than 10 minutes"

Note: Audio-only streams need only comply with App Store Review Guidelines 5.1.2.

"Audio streaming content over a cellular network may not use more than 10 minutes of audio."

When streaming video content over a **cellular** network with a duration longer than 10 minutes, you **must provide a baseline stream with a duration of 10 minutes or less**.

Note: If you cannot provide video of acceptable quality at 192 kbps or less, you must provide a baseline stream with a duration of 10 minutes or less.

The image shows the Apple logo at the top left. Below it, a collection of Apple devices is displayed, all showing a live stream of a person wearing a racing helmet and goggles. The devices include a large iMac monitor, a laptop, a smartphone, and two tablets. The person in the helmet is wearing an orange and white racing suit. The background of the stream is dark with some mechanical parts visible.

HLS Playback Device Support

Desktop Browser Support

✔ supported

✘ not supported

⚙️ we are working on it right now. [Join our Newsletter](#) to stay up to date with the latest updates!

Device	DASH HTML5	DASH Flash	HLS HTML5	HLS Flash
Chrome	✔ ¹	✔	✔ ¹	✔
Firefox	✔ ²	✔	✔ ²	✔
Internet Explorer 9+	✔ ³	✔	✔ ³	✔
Microsoft Edge	✔	✔	✔	✔
Safari	✔ ⁴	✔	✔ ⁴	✔
Opera	✔ ⁵	✔	✔ ⁵	✔

Streaming Devices

✔ supported

✘ not supported

⚙️ we are working on it right now. [Join our Newsletter](#) ¹

Device	DASH HTML5	HLS HTML5
Chromecast	✔	✔
Apple TV	✘	⚙️
Amazon FireTV	✔	✔



Mobile Devices

✔ supported

✘ not supported

⚙️ we are working on it right now. [Join our Newsletter](#) to stay

Device	DASH HTML5	HLS HTML5
Chrome on Android ¹	✔	✔
Firefox on Android ²	✔	✔
Safari on iOS ³	✘	✔
Windows Phone ⁴	✔	✘

TVs

✔ supported

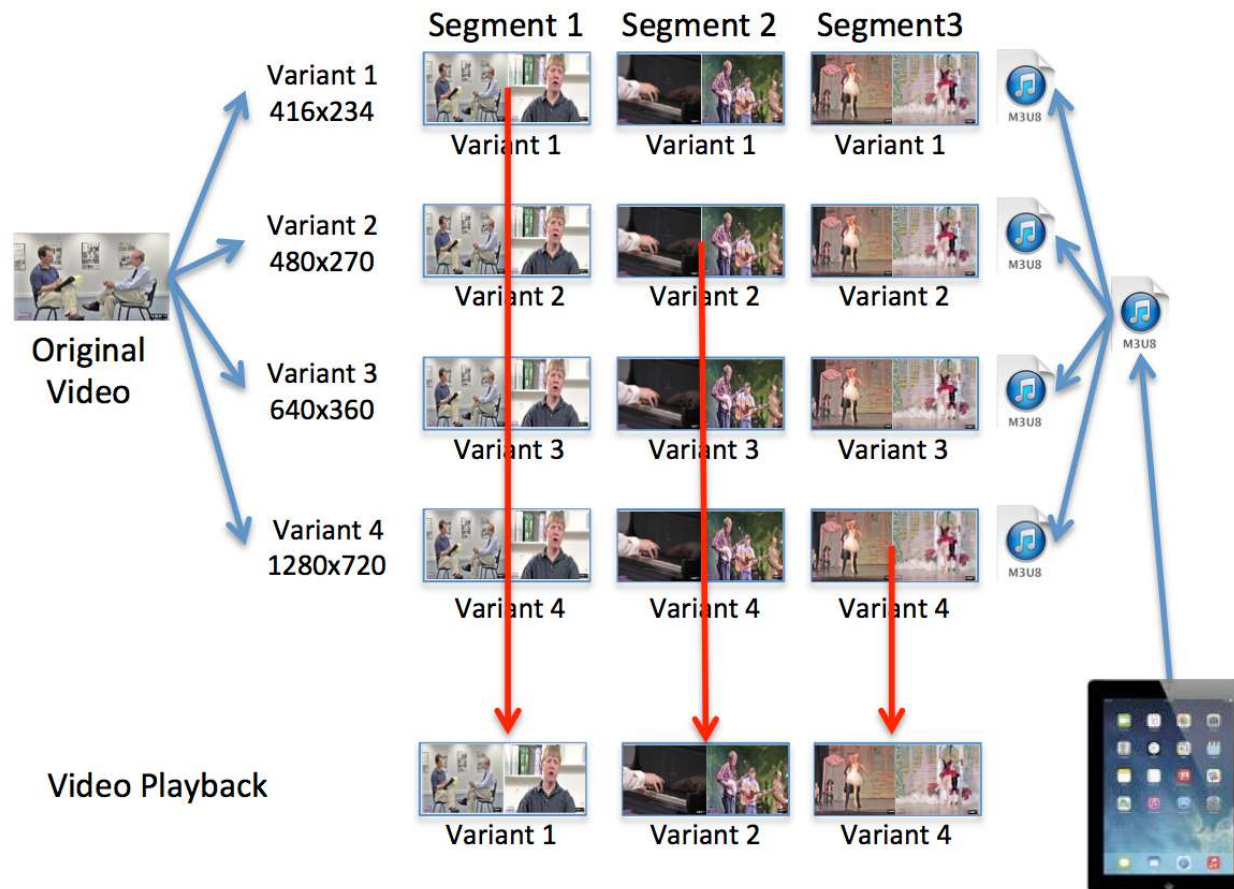
✘ not supported

⚙️ we are working on it right now. [Join our Ne](#)

Device	DASH HTML5	HLS HTML5
HbbTV ¹	✔	✔

HLS Playback Device Bitrates

Adaptative Bitrate -> Simultaneous Multiple Bitrates (Variants)



HLS Playback Device Bitrates



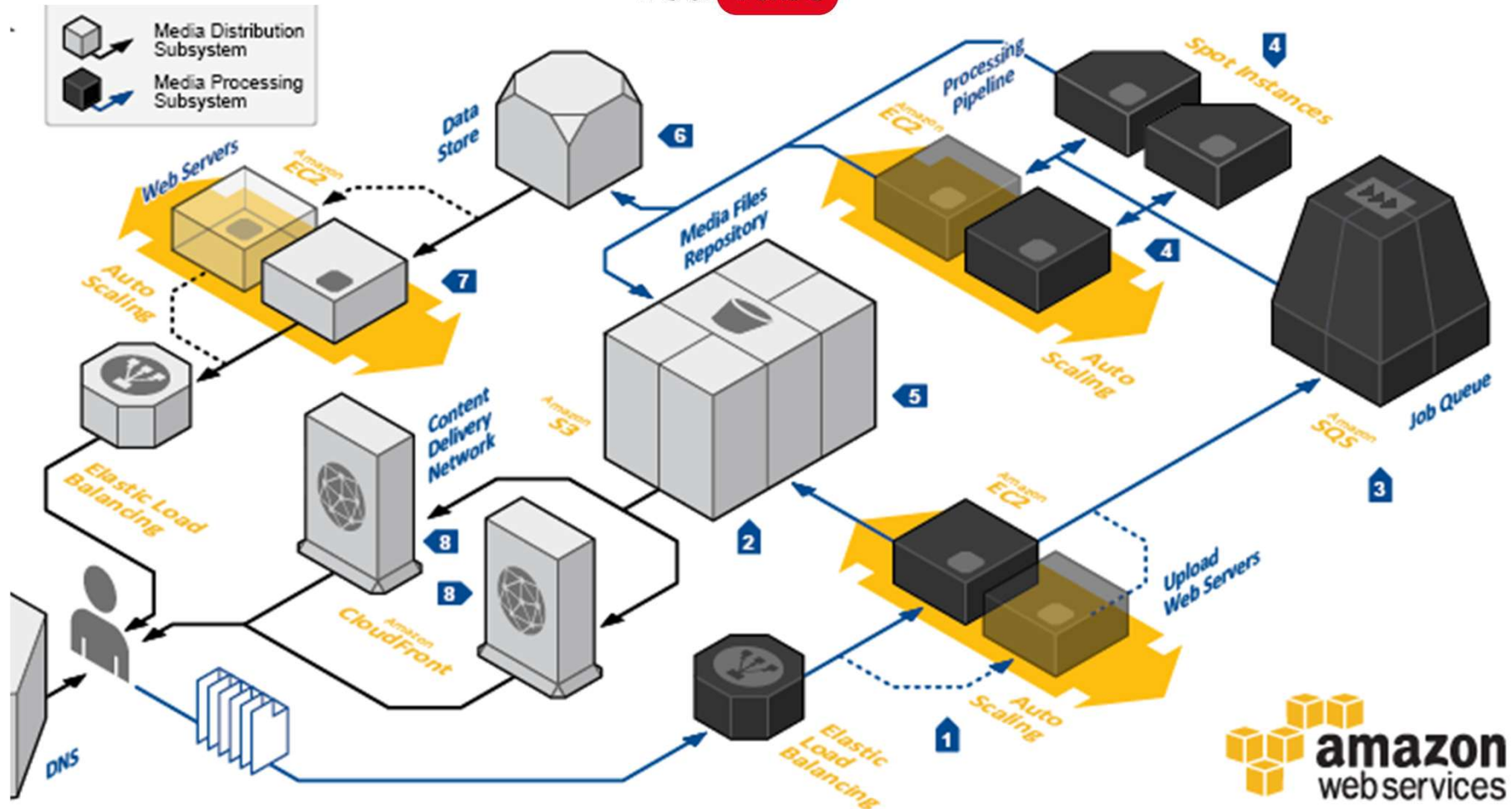
Clients			Dimensions for 16:9 aspect ratio	Dimensions for 4:3 aspect ratio	Frame rate	Video bit rate (average)	Video bit rate (peak)	Audio bit rate	Total bit rate
	CELL		416 x 234	400 x 300	12	145	200	64	264
	CELL	ATV	480 x 270	480 x 360	15	365	400	64	464
WiFi	CELL	ATV	640 x 360	640 x 480	29.97	730	800	64	864
WiFi	CELL	ATV	768 x 432	640 x 480	29.97	1100	1200	96	1296
WiFi		ATV	960 x 540	960 x 720	29.97 or source	2000	2200	96	2296
WiFi		ATV	1280 x 720	960 x 720	29.97 or source	3000	3300	96	3396
WiFi		ATV	1280 x 720 or source	1280 x 960 or source	29.97 or source	4500	5000	128	5128
WiFi		ATV	1280 x 720 or source	1280 x 960 or source	29.97 or source	6000	6500	128	6628
WiFi		ATV	1920 x 1080	1920 x 1440	29.97 or source	7800	8600	128	8728

Digital Streaming Architectures

(Amazon Web Services)

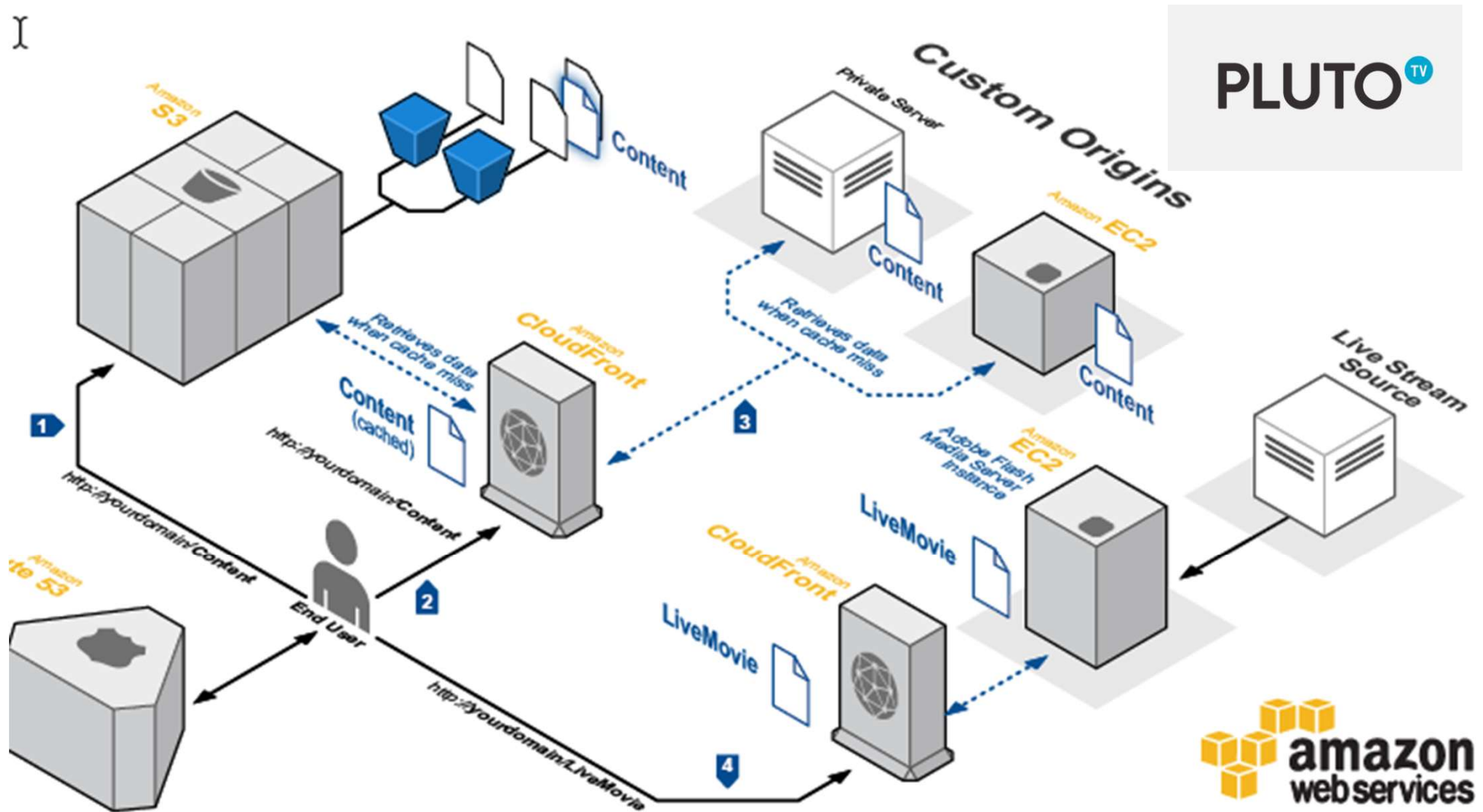
Media Sharing Architecture

You Tube



- Bandwidth vs. Stream Transfer
- Security
- Compute (Encoding?)
- Storage

Media Serving Architecture



- Bandwidth vs. Stream Transfer
- Security
- Compute (Encoding?)
- Storage

Don't Forget Security

- **Hot-linking** protection.
- **Re-streaming** protection.
- **Domain locking** geo-location and IP-based.
- **Digital Right Management / DRM** (Google Wildvine, Microsoft Playready, Adobe Flash Access, Apple Fairplay).

Streaming Solutions

- **Services** (Content Delivery Network / CDN)
 - **Akamai**, Amazon Web Services (Elastic Transcoder / CloudFront), Azure Media Services, bitmovin, **Ustream**, Limelight Network, Tata Communications, Brightcove Zencoder, **encoding.com**, Altitude Multicast+, smoothHD.com
- **Servers** (Streaming Server / SS)
 - **Adobe Media Server**, Apache, Elemental Technologies, Helix Universal Server, Darwin SS, **Nginx**, **Nimble Streamer**, Red5, Unified Origin, Unreal Media Server, Microsoft IIS w/Smooth Streaming Extension, **Wowza Streaming Engine**

Contactos

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